Unit 3, Lesson 4: Dividing Rational Numbers

Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

Learning Target

☐ I can fluently divide rational numbers.

3.4.1 Warm-Up: Would You Rather?

1. Explain if you would rather have a stack of quarters that stands from the floor to the top of your head, or $\$225.00$.

2. Explain whether you would change your mind if the quarters were stacked to the top of your desk.

3.4.2 Refresher: Dividing Positive Rational Numbers

1. Ren found the following quotient, making at least one error in her work.

   a. Circle the error(s) in Ren's work and write an explanation of the error(s). Then, find the correct quotient.

Ren's Work	Explanation	Correct Quotient
$12.636 \div 3.12$ $\begin{array}{r} .0\,4\,0 \\ 3.1\,2\overline{)1\,2.6\,3\,6} \\ -1\,2\;\;4\,8 \\ 1\,5\,6 \\ -\quad 0 \\ 0 \\ \hline \end{array}$		

   b. What advice do you have for Ren that could help her with dividing decimals in the future?

2. Trevor found the following quotient, making at least one error in his work. Circle the error(s) in Trevor's work. Write an explanation of the error(s). Then, find the correct quotient.

Trevor's Work	Explanation	Correct Quotient
$\left(\frac{38}{13}\right) \div \left(1\frac{6}{13}\right)$ $\left(\frac{38}{13}\right) \div \left(\frac{19}{13}\right)$ $\frac{(38 \div 19)}{13}$ $\frac{2}{13}$		

3.4.3 Collaboration: Signed Number Rules

1. Fill in the blanks with the words *positive* or *negative* to write the rules for dividing integers.

   - The quotient of two positive integers is a _________ integer.
   - The quotient of two negative integers is a _________ integer.
   - The quotient of a negative integer divided by a positive integer is a _________ integer.
   - The quotient of a positive integer divided by a negative integer is a _________ integer.

2. Discuss with your partner whether you think the same rules apply to rational numbers. Write a summary of your reasoning.

3. Complete the table. Use a calculator to find the quotient of each expression to verify if the same rules for signs apply when dividing rational numbers.

Expression	Sign of Quotient Using the Rules	Quotient	Did the Same Rule Apply?
$\left(-2\frac{3}{8}\right) \div \left(-1\frac{1}{4}\right)$			
$6\frac{7}{18} \div -4\frac{2}{9}$			
$43.008 \div -4.096$			
$(-525.6) \div (-0.36)$			

4. Complete the sentence.

   The rules for dividing rational numbers are ○ the same as ○ different from the integer rules for division.

3.4.4 Guided Practice: Finding the Quotients of Rational Numbers

1. Circle the expressions that will result in a positive quotient.

$^-9.655 \div 0.5$ $\quad$ $0.068 \div {}^-0.01$ $\quad$ $^-14.628 \div {}^-4.6$

$\frac{16}{20} \div \left(-\frac{8}{15}\right)$ $\quad$ $\left(-\frac{9}{6}\right) \div \left(-\frac{5}{3}\right)$ $\quad$ $\left(-\frac{3}{5}\right) \div 2\frac{1}{2}$

2.

a. Write a division sentence in the form of $a \div b$ where the value of a is a number **between** 15.1 and 16.2, and b is a number **between** $^-4.6$ and $^-3.3$.

b. What is the value of the quotient?

3.4.5 Your Turn!

1. Find the value of each expression.

a. $\left(-3\frac{5}{9}\right) \div \left(3\frac{1}{3}\right)$

b. $70.62 \div {}^{-}160.5$

c. $\left(-1\frac{6}{7}\right) \div \left(-5\frac{3}{4}\right)$

d. ${}^{-}44.64 \div {}^{-}2.88$

e. $8\frac{1}{5} \div \frac{3}{2}$

f. $-45.018 \div (-7.32)$

3.4.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to each statement. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can correctly decide the sign of the quotient when dividing rational numbers.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.
←		→

Evidence:

2. I can fluently divide rational numbers that are fractions.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.
←		→

Evidence:

3. I can fluently divide rational numbers that are decimals.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.
←		→

Evidence: